Medical Image Classification for Early Melanoma Detection

Bryan Torres
Virginia Tech
Department of Computer Science
bryantorres@vt.edu

Roman Chenoweth
Virginia Tech
Department of Computer Science
romanc22@vt.edu

Cian Teague
Virginia Tech
Department of Computer Science
cianteague@vt.edu

Abstract

Melanoma is a serious form of skin cancer where early detection can substantially improve outcomes. This project studies binary classification of dermoscopic lesion images into benign and malignant categories using transfer-learned convolutional neural networks. Using an ISIC-derived Kaggle dataset with 13,879 total images, we built a complete image-classification pipeline including stratified train/validation splitting, preprocessing, data augmentation, two-phase fine-tuning, and evaluation across five pretrained CNN architectures. The final benchmark shows strong performance across all models, with ResNet-50 achieving the best overall result: AUC-ROC of 0.9800, accuracy of 0.9185, F1 score of 0.9154, and sensitivity of 0.9850 at 80% specificity on a balanced held-out test set. Overall, the project demonstrates that a carefully implemented transfer-learning pipeline can produce strong melanoma-screening performance while also showing why reliable splitting, clinical metrics, and responsible interpretation are essential for medical-image machine learning.

1 Introduction and Motivation

Melanoma is a high-risk skin cancer, and early detection is one of the most important factors in improving patient outcomes. In practice, visual inspection of skin lesions can be difficult because benign and malignant lesions may share similar visual patterns. Dermoscopic images also vary in illumination, zoom, lesion size, skin tone, and artifact content such as hair, rulers, shadows, and dark borders. These issues make melanoma detection a meaningful machine learning problem, especially when the goal is to support screening rather than replace clinical judgment. The motivation for this project is to study whether modern deep neural networks can learn useful visual representations for distinguishing malignant melanoma from benign lesions. We frame the system as a decision-support prototype. In this setting, the most important concern is not simply maximizing accuracy, but reducing missed malignant cases while still maintaining a reasonable specificity. This means the project has asymmetric error costs: a false negative is usually more concerning than a false positive because it may represent a missed cancer case. From a technical perspective, this project is a supervised binary image-classification task. The main work involved building a reliable training pipeline, comparing multiple pretrained CNN backbones, and evaluating the models with clinically meaningful metrics. The final project focuses on whether transfer learning, image augmentation, and threshold-based evaluation can produce strong and interpretable results on a balanced held-out dermoscopy test set.

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2 Problem Definition

Let an input dermoscopic image be represented as

$$x \in \mathbb{R}^{224\times 224\times 3},$$

where each image is resized to 224 by 224 pixels with three RGB channels. Each image has a binary label

$$y \in \{0, 1\},$$

where $y = 0$ denotes benign and $y = 1$ denotes malignant. The goal is to learn a model

$$f_\theta(x) \in [0, 1]$$

that estimates the probability that the lesion is malignant. Given a decision threshold τ, the predicted label is

$$\hat{y} = \mathbb{I}[f_\theta(x) \geq \tau].$$

The learning problem is therefore to find parameters θ that separate benign and malignant lesions well on unseen images. Because this is a medical-screening style task, success is measured using more than raw accuracy. We evaluate AUC-ROC, precision, recall, F1 score, Brier score, and sensitivity at fixed specificity values. Sensitivity at fixed specificity is especially important because it shows how many malignant lesions the model can detect while holding the benign false-positive rate to a controlled level.

3 Related Work and Survey

Deep learning has become a common approach for skin lesion analysis because convolutional neural networks can learn hierarchical visual features directly from image data. Esteva et al. (1) showed that deep CNNs trained with transfer learning can achieve dermatologist-level classification performance on curated skin lesion datasets. This paper is important for our project because it supports the core idea of using ImageNet-pretrained networks and fine-tuning them for medical image classification. Haenssle et al. (2) compared a deep CNN against dermatologists for dermoscopic melanoma recognition. A useful part of that study is its clinical framing: the model is evaluated at specific sensitivity and specificity operating points rather than only through overall accuracy. This influenced our decision to report sensitivity at 80% and 90% specificity. The HAM10000 dataset introduced by Tschandl et al. (3) and the ISIC challenge benchmark summarized by Codella et al. (4) helped standardize skin lesion research. These datasets made it easier to compare models, but they also highlight common medical-image issues such as class imbalance, limited metadata, duplicate-like samples, and possible dataset bias across imaging conditions and patient groups. For model design, ResNet (5), DenseNet (6), EfficientNet (7), and MobileNetV2 (8) offer different trade-offs. ResNet uses residual connections to train deeper networks, DenseNet reuses features through dense connections, EfficientNet scales network depth, width, and resolution in a balanced way, and MobileNetV2 is designed for lighter deployment. Since these architectures have different strengths, our project compares all of them under the same data split, preprocessing, augmentation, and evaluation procedure. Our regularization and imbalance-handling choices are also based on prior work. Focal loss (9) reduces the influence of easy examples and focuses learning on harder cases. MixUp (10) creates soft training examples by interpolating between images and labels, while CutOut (11) masks random regions of an image to prevent the model from depending too heavily on one visual cue. These methods are useful in our project because medical images can contain artifacts or local patterns that may cause overfitting.

4 Proposed Method

4.1 Intuition

Our method is based on the idea that pretrained CNNs already contain useful low-level and mid-level visual features, such as edges, textures, colors, and shapes. Instead of training a deep network from scratch, we use transfer learning to adapt these features to dermoscopic lesion classification. This is helpful because medical datasets are usually smaller than natural-image datasets, so training from scratch would be more likely to overfit. The project also uses a controlled comparison. Rather than

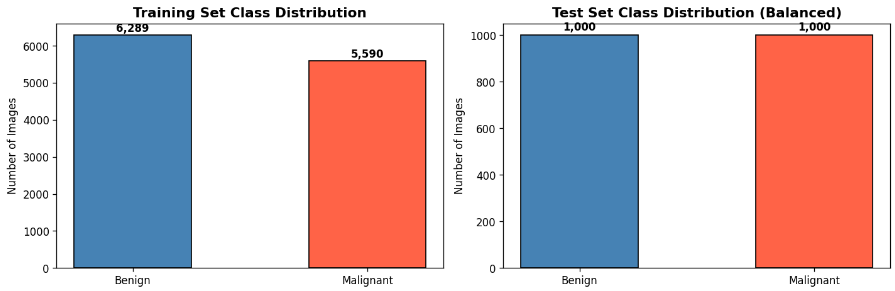

Figure 1: Class distribution for the packaged dataset. The training set is mildly imbalanced, while the held-out test set is balanced.

changing several parts of the pipeline at once, we keep the classification head, data split, augmentation procedure, and evaluation process consistent across architectures. This makes the comparison more meaningful because differences in performance are more likely to come from the backbone itself rather than from unrelated training choices.

4.2 Dataset and Preprocessing

We used the "Skin Cancer: Malignant vs. Benign" dataset derived from the ISIC archive and hosted on Kaggle. The dataset is provided as a directory-structured image collection, with RGB dermoscopic images stored in JPEG format. Every image is resized to 224×224 pixels so that it can be used with standard ImageNet-pretrained architectures. The packaged dataset contains 13,879 total images. The provided training directory contains 11,879 images, with 6,289 benign images and 5,590 malignant images. The held-out test directory contains 2,000 images, evenly balanced with 1,000 benign and 1,000 malignant images. From the provided training directory, we created a stratified 90/10 train/validation split. This resulted in 10,692 training images and 1,187 validation images. The training split contains 5,661 benign and 5,031 malignant examples, while the validation split contains 628 benign and 559 malignant examples. One important implementation issue involved validation splitting. Our first version used automatic directory loading utilities, but because filenames were grouped alphabetically by class, the validation slice could accidentally contain mostly one class. This caused unreliable validation metrics and made early stopping untrustworthy. To fix this, we manually collected files per class and created a stratified shuffle split. This correction was one of the most important parts of the project because the final benchmark depends on a valid validation set.

4.3 Model Architectures

We benchmarked five pretrained CNN architectures: EfficientNet-B0, EfficientNet-B2, ResNet-50, DenseNet-121, and MobileNetV2. Each model used the same classification head:

$$\texttt{GlobalAveragePooling} \rightarrow \texttt{Dense(256, ReLU)} \rightarrow \texttt{BatchNorm} \rightarrow \texttt{Dropout(0.3)} \rightarrow \texttt{Dense(1, sigmoid)}.$$

The sigmoid output gives the predicted probability that the lesion is malignant. All architectures were trained using the same two-phase fine-tuning strategy. In the first phase, the pretrained backbone was frozen and only the classification head was trained. In the second phase, the backbone was unfrozen and fine-tuned end-to-end using a lower learning rate. Batch normalization layers remained frozen during fine-tuning to preserve stable pretrained statistics.

4.4 Losses, Augmentation, and Regularization

The main benchmark used weighted binary cross-entropy. For a predicted probability $p = f_\theta(x)$ and label y, the loss is

$$\mathcal{L}_{\text{WCE}} = -w_1 y \log p - w_0(1-y)\log(1-p),$$

where w_0 and w_1 are class weights derived from the training split. Since the dataset version is only mildly imbalanced, the class weighting is not extreme, but it still helps account for the slightly smaller malignant class. We also considered focal loss as an imbalance-focused alternative:

$$\mathcal{L}_{\text{FL}} = -\alpha_t(1-p_t)^\gamma \log(p_t),$$

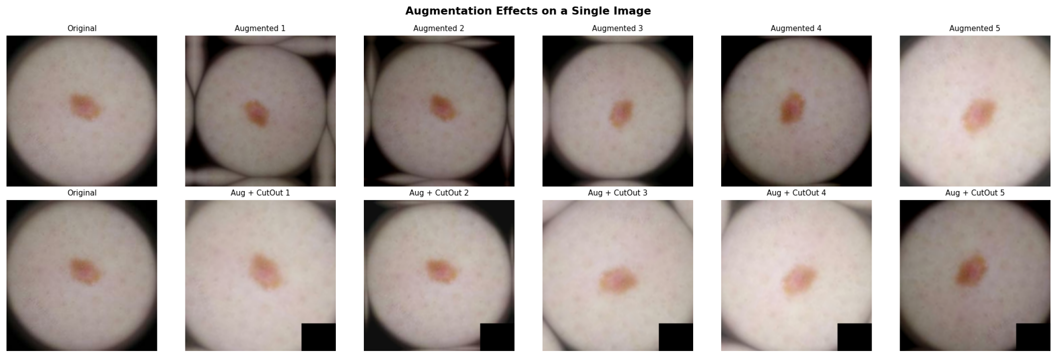

Figure 2: Examples of the augmentation pipeline applied to a lesion image. MixUp is applied at the batch level, while CutOut masks a random region of an image during training.

Table 1: Architecture benchmark on the balanced held-out test set. All models used the same preprocessing, augmentation pipeline, two-phase fine-tuning procedure, and weighted cross-entropy setup.

Architecture	Accuracy	AUC	Precision	Recall	F1	Sens.@80% Spec	Sens.@90% Spec	Brier
EfficientNet-B0	0.9090	0.9748	0.9495	0.8640	0.9047	0.9790	0.9480	0.0763
EfficientNet-B2	0.9090	0.9743	0.9398	0.8740	0.9057	0.9830	0.9500	0.0768
ResNet-50	**0.9185**	**0.9800**	**0.9515**	0.8820	**0.9154**	**0.9850**	**0.9600**	**0.0666**
DenseNet-121	0.9130	0.9732	0.9366	**0.8860**	0.9106	**0.9850**	0.9450	0.0748
MobileNetV2	0.9030	0.9713	0.9400	0.8610	0.8987	0.9830	0.9400	0.0808

where p_t is the probability assigned to the true class, α_t is a class-balancing term, and γ controls how much the loss focuses on hard examples. In our final report, we treat focal loss as an important extension and analysis point because increasing γ can make the model pay more attention to uncertain or misclassified images. The augmentation pipeline included random horizontal and vertical flips, full-range random rotation, random zoom, brightness and contrast perturbations, CutOut, and MixUp. MixUp creates soft training examples:

$$\tilde{x} = \lambda x_i + (1 - \lambda) x_j, \qquad \tilde{y} = \lambda y_i + (1 - \lambda) y_j.$$

CutOut masks a random square region of the image. Together, these augmentations encourage the model to learn more general lesion patterns instead of memorizing artifacts or small local details.

5 Experiments and Results

5.1 Experimental Questions and Testbed

The experiments were designed to answer four main questions. First, can transfer-learned CNNs achieve strong performance on benign versus malignant dermoscopic image classification? Second, which pretrained architecture performs best under a consistent training and augmentation pipeline? Third, do the models maintain high sensitivity at clinically meaningful specificity levels? Fourth, what implementation issues affect the reliability of the benchmark? All models were trained on the stratified training split and selected using validation loss. Final evaluation was performed only on the balanced held-out test set. Predictions were exported as probabilities and evaluated externally using scikit-learn. We computed accuracy, AUC-ROC, precision, recall, F1 score, Brier score, and sensitivity at 80% and 90% specificity.

5.2 Architecture Benchmark

Table 1 shows the final architecture benchmark. All five models achieved AUC above 0.97, which suggests that transfer learning works well for this dataset. ResNet-50 had the best overall performance, with the highest AUC, accuracy, F1 score, sensitivity at 90% specificity, and lowest Brier score.

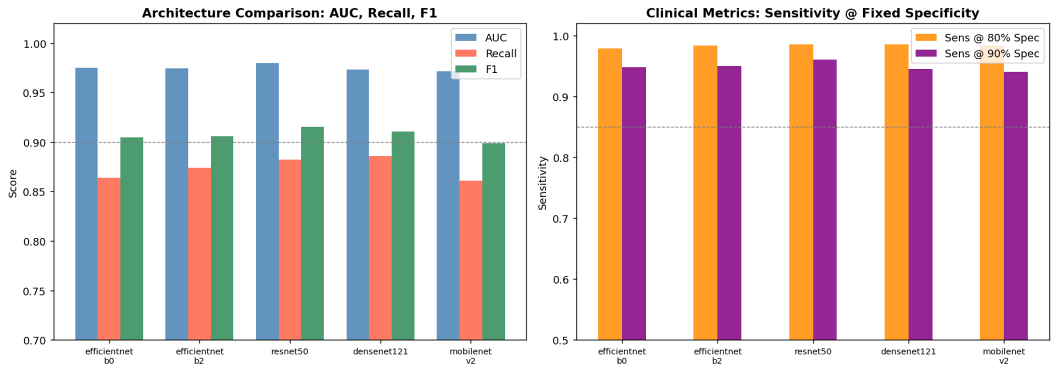

Figure 3: Comparison of AUC, recall, F1, and sensitivity at fixed specificity across the five benchmarked architectures.

5.3 Observations and Interpretation

The first main observation is that ResNet-50 was the strongest all-around model. It achieved an AUC of 0.9800 and the lowest Brier score of 0.0666, which means it performed well both as a ranking model and as a probability-producing model. Its sensitivity at 90% specificity was also the best among the models, which matters because melanoma screening needs high malignant-case detection while still limiting unnecessary false positives. The second observation is that DenseNet-121 was very competitive. It had the highest thresholded recall at the default decision threshold, with recall of 0.8860 compared with ResNet-50's 0.8820. This difference is small, but it shows that model selection depends on the chosen operating point. If the main goal is default-threshold recall, DenseNet-121 is close to ResNet-50. If the main goal is overall discrimination, calibration, and sensitivity at 90% specificity, ResNet-50 is stronger. The third observation is that MobileNetV2 performed slightly worse but still achieved an AUC of 0.9713. This makes it a reasonable deployment-friendly option if the project were extended into a lightweight web demo or mobile setting. However, because this project focuses on classification performance rather than deployment speed, ResNet-50 is the final preferred model. The final observation is that the benchmark depends heavily on correct data handling. The validation split issue showed that a model can appear unstable or misleading if the validation set is not properly stratified. After correcting the split, the results became much more interpretable and consistent.

5.4 Focal Loss Gamma Discussion

The proposal feedback suggested that we analyze the focal loss focusing parameter γ. In this project, γ is important because it controls how strongly focal loss down-weights easy examples. A smaller value behaves more like weighted binary cross-entropy, while a larger value focuses more on hard or misclassified cases. In a melanoma-screening setting, this could be useful because difficult malignant examples are clinically important. For the final system, we report the weighted cross-entropy benchmark as the main result because it was stable, easy to interpret, and already produced strong performance across all architectures. Focal loss remains a useful extension for future work, especially if the dataset becomes more imbalanced or if the goal shifts toward maximizing sensitivity at a stricter specificity level.

6 Software Package and Reproducibility

The final project code, notebook workflow, and generated result files are maintained in a GitHub repository so that the project can be reviewed, reproduced, and extended. The repository contains the main exploration notebook, supporting Python code, saved benchmark outputs, and figures used in the final report.

6.1 Repository and Package Information

GitHub repository:
`https://github.com/RomanLearns/melanoma`

Submitted package file:
`[Fill in submitted tar.gz file name here.]`

Notebook entry point:
`capstone_exploration.ipynb`

Supporting code file:
`cell_fixes.py`

6.2 Current Repository Structure

The final project repository and submitted package are organized as follows:

```
melanoma/
  README.txt
  002.zip
  DOC/
    final_report.pdf
    final_report.tex
    graded_proposal.pdf
    graded_milestone.pdf
  SRC/
    capstone_exploration.ipynb
    cell_fixes.py
    capstone_results/
      arch_benchmark.pkl
      arch_benchmark.png
      augmentation_examples.png
      class_distribution.png
      sample_images.png
```

The `README.txt` file serves as the user manual for the project and explains the project goal, required packages, repository structure, dataset, dataset setup, and how to run the notebook workflow. The `DOC/` directory contains the final report materials and copies of the graded proposal and milestone documents. The `SRC/` directory contains the main project notebook, `capstone_exploration.ipynb`, the supporting implementation file `cell_fixes.py`, and the generated result files used in the report. The `capstone_results/` directory stores the saved benchmark results and figures, including the architecture benchmark plot, augmentation examples, class distribution figure, and sample lesion images.

7 Difficulties, Limitations, and Ethical Considerations

The main technical difficulty was the validation split failure caused by directory-order loading. This was an important lesson because the model itself was not the only possible source of error. A flawed data pipeline can make validation results unreliable even if the architecture and training code are correct. Fixing the split with a manual stratified shuffle was necessary before the benchmark could be trusted. A second difficulty was framework metric reliability. In repeated experiments, relying too heavily on framework-side validation metrics can be risky because stateful metric objects may behave unexpectedly across repeated model compilation. To reduce this risk, final metrics were computed outside the training loop using scikit-learn. This project also has medical-data limitations. The dataset is useful for an academic benchmark, but we do not have enough metadata to evaluate fairness across skin tones, age groups, imaging devices, or clinical sites. Because of this, the model should not be interpreted as a deployable clinical diagnostic tool. It is a research prototype that demonstrates image-classification methods, not a replacement for a dermatologist or medical professional. Any future version of this system should include stronger external validation, fairness analysis, clinical review, calibration testing, and clearer uncertainty reporting. A real medical deployment would also require privacy safeguards, regulatory review, and careful human oversight.

8 Conclusion and Future Work

This project built a complete deep learning pipeline for binary melanoma image classification. We prepared an ISIC-derived Kaggle dataset, corrected an important validation-splitting issue, applied a consistent augmentation pipeline, trained five pretrained CNN architectures, and evaluated the models on a balanced held-out test set. The strongest final result came from ResNet-50, which achieved an AUC-ROC of 0.9800, accuracy of 0.9185, F1 score of 0.9154, and sensitivity of 0.9850 at 80% specificity. The main conclusion is that transfer learning is effective for this melanoma-classification benchmark, but the reliability of the result depends heavily on careful preprocessing, stratified splitting, and clinically meaningful evaluation metrics. The project also shows that different architectures offer different trade-offs. ResNet-50 was the best overall model, DenseNet-121 was competitive for recall, and MobileNetV2 remained useful as a lighter deployment candidate. Future work could extend this project by running a more complete focal-loss and oversampling ablation, adding Grad-CAM visualizations for interpretability, evaluating calibration more deeply, and testing the model on external datasets. The most important future step would be external validation across more diverse patient groups and imaging conditions before making any stronger claims about real-world medical usefulness.

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